PHYS281 Report on an Astronomical Simulation

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Abstract

In this report we will describe how our N-body gravitational simulation of the Solar System can show both the accuracies and limitations of a variety of different integrating algorithms, by comparing the results of the simulation between them. We will show how decreasing the values of time steps in these algorithms can improve accuracy, and how the algorithms respond over larger time ranges, as well as showing the accuracy of the simulation with regards to Energy and Momentum Conservation, and Kepler's 3rd Law. We show that the Verlet Method of integration works best out of the three methods at larger time ranges and small time steps, giving a value of 365.080 for the length of an Earth year in days, compared to 319.405 for the Euler-Cromer method, and 394.679 for the Euler Method at the same scale. The Verlet Method also prevails for all the tests over longer time steps, consistently conserving a total linear energy of $< 0.01\%$, compared to 2.24 % for the Euler method at the largest time step of 500,000, and 0.03 % for the Euler-Cromer Method under the same conditions (by referring to NASA ephemeris data at the end point of the simulation). Furthermore, we show how the Euler and Euler-Cromer Methods retain a very good level of accuracy at smaller time steps, and smaller scales, but do have a greater error accumulate as the number of iterations increases. We conclude that the Verlet Method is the most suitable of the three for more complex simulations, and further explain the significance of ephemeris data when calculating % errors.

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1 Introduction

Much of the modern space industry, Astronomy, and Astrophysics rely on accurate simulations of orbital mechanics. Thus, improving the accuracy of these simulations is of paramount importance for much of modern research. This report will compare three main algorithms for a gravitational simulation: the Euler Method, the Euler-Cromer Method, and the Verlet Method. By comparing the time periods of planets in the simulation, alongside whether the simulation accurately abides Energy and Momentum Conservation the accuracy of each algorithm is assessed. From this, we propose ways to further improve precision, as well as to discuss any limitations in the code.

Using ephemeris data from NASA's JPL website, each object in the simulation was given accurate initial position and velocity vectors. From there, the acceleration of each object was updated using the aforementioned algorithms, which in turn updated the position and velocity vectors over a set time range, and in successive set time intervals. Using Python's MatPlotLib library, this data was plotted in a 3D graph (to more easily observe the orbits' stability and other inconsistencies), alongside 2D graphs of the total linear and angular kinetic energy, and of the linear and angular momenta in the simulation. The time periods of set objects was also generated using Kepler's 3rd Law, which could be compared to accurate ephemeris data. This was done for each algorithm and the differences tabulated,

Background Theory:

To update the acceleration each time, a rearranged form of Newton's Law of Gravitation was used [1]:

$$\vec{a_i} = \sum_{j \neq i}^{N} \frac{-Gm_j}{|\vec{r_{ij}}|^2} \, \hat{\vec{r}}_{ij}$$

Where $\vec{a_i}$ denotes the acceleration of the ith particle, G is the Gravitational constant, and $|\vec{r_{ij}}|^2 = (x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2$, which is the magnitude of the differences in the positions of the i_{th} and j_{th} particles.

The three algorithms included:

The Euler Method:

$$\vec{R}_{n+1} \approx \vec{R}_n + \vec{V}_n \Delta t$$
$$\vec{V}_{n+1} \approx \vec{V}_n + \vec{A}_n \Delta t$$

Where $\vec{R}_n$ denotes the nth term of the position vector, $\vec{V}_n$ denotes the nth term of the velocity vector, $\vec{A}_n$ denotes the nth term of the acceleration vector and Δt denotes the time steps in the algorithm. The Euler Method is based on the Taylor Series expansion of the

standard equations of motion, which also gives an extra, unincluded term of $O(\Delta t^2)$[2]. Hence, an error of $N\Delta t^2$ accumulates for every N^{th} iteration of this method. This causes the Euler Method to be the least accurate of all the approaches used in the program, but also the quickest to compute.

The Euler-Cromer Method:

$$\vec{V}_{n+1} \approx \vec{V}_n + \vec{A}_n \Delta t$$
$$\vec{R}_{n+1} \approx \vec{R}_n + \vec{V}_{n+1} \Delta t$$

Where $\vec{R}_n$ denotes the nth term of the position vector, $\vec{V}_n$ denotes the nth term of the velocity vector, $\vec{A}_n$ denotes the nth term of the acceleration vector and Δt denotes the time steps in the algorithm, just as earlier. The Euler-Cromer Method, as the name suggests, is simply a modification of the Euler Method, but with the velocity at the beginning of the step. It has a rather interesting history, having been seemingly lost and rediscovered many times over the centuries[3]; most recently being "discovered" by a High School student in 1980 who accidentally rewrote Basic code the wrong way around[4]. Just as with the standard Euler Method, an error of $N\Delta t^2$ still accumulates for every N^{th} iteration of the algorithm, but it is much better at conserving energy in the simulation[5], as shown later.

The Verlet Method:

$$\vec{R}_{n+1} \approx \vec{R}_n + \vec{V}_n \Delta t + \tfrac{1}{2}\vec{A}_n \Delta t^2$$
$$\vec{V}_{n+1} \approx \vec{v}_n + \tfrac{1}{2}(\vec{A}_{n+1} + \vec{A}_n)$$

Where $\vec{R}_n$ denotes the nth term of the position vector, $\vec{V}_n$ denotes the nth term of the velocity vector, $\vec{A}_n$ denotes the nth term of the acceleration vector and Δt denotes the time steps in the algorithm, just as in the previous two methods.

The Verlet method is a more accurate alternative to the previous two methods by using the end acceleration to calculate an updated velocity. This helps smooth out any changes in the acceleration. Interestingly, this algorithm can be derived by successively alternating between the previous two methods[6], which results in it having an error of $N\Delta t^4$ instead[7], a whole order of magnitude greater accuracy than the previous methods. The repercussions of this in Energy and Momentum conservation, as well as when comparing the approximations of time periods, will be discussed in more depth later.

2 Validation of methodology

The simulation can accurately generate 3D graphs of the orbits of the bodies of the Solar System over variable time ranges. Some of the produced graphs have been added below to

validate that the program functions correctly. All of these were determined using a time step of 50,000 seconds, from the 26th December 2004, using the Verlet Method of integration.

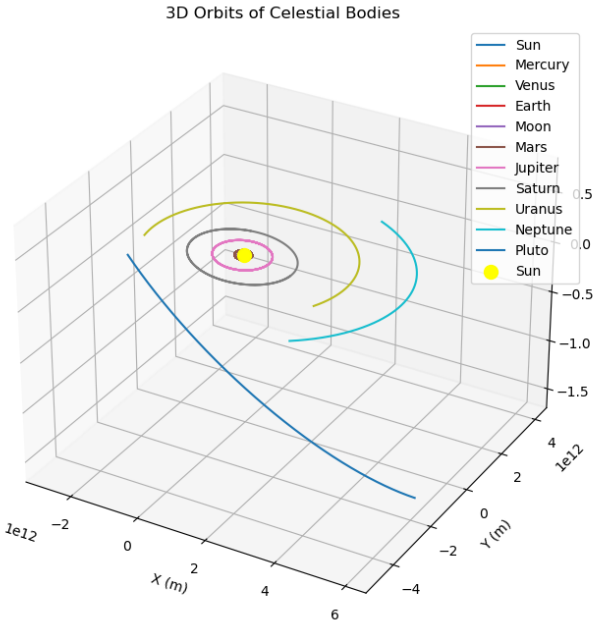


Figure 1: 3D graph of the Solar System with a time range of 50 years. Notice the incomplete orbits of the outermost planets (due to the smaller time range).

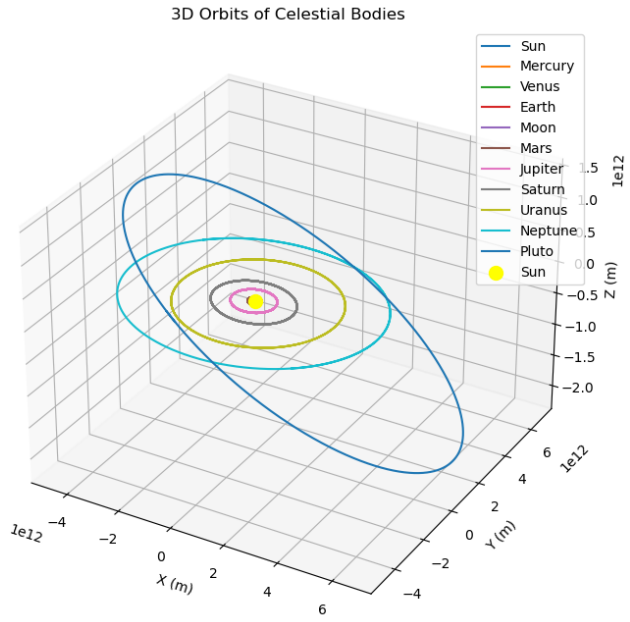


Figure 2: A contrasting graph of the Solar System with a time range of 250 years. Notice how all the orbits are complete, as expected.

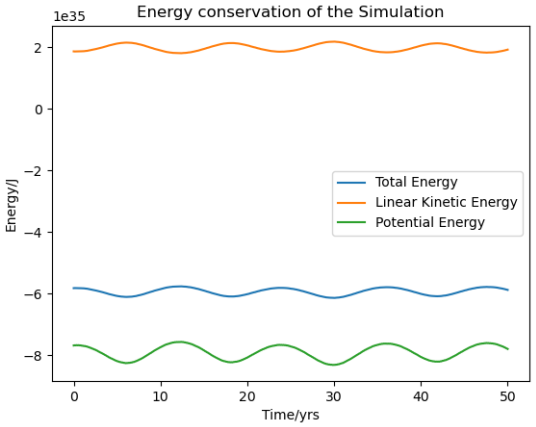


Figure 3: A graph that shows the Energy conservation of the system over 50 years. Notice how E_{total} is negative, and how the +KE and -PE oscillate at the same rate, but in opposing directions.

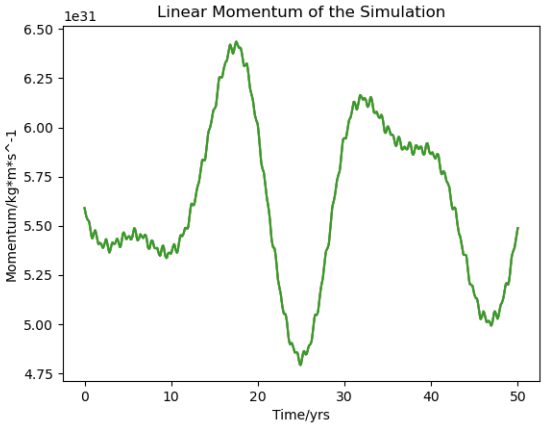


Figure 4: A graph that shows the linear momentum conservation of the system over 50 years. Notice the roughness of this graph compared to the others due as the orbits of each body in the system change.

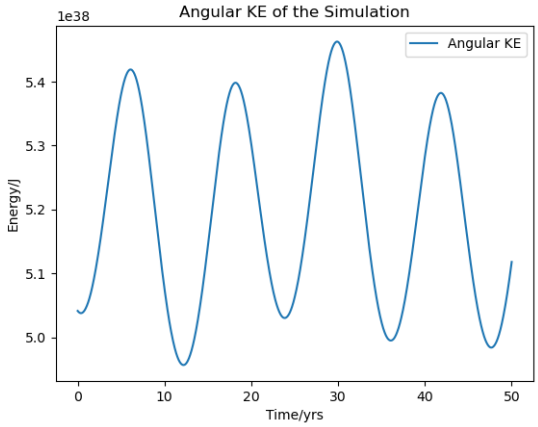


Figure 5: A 2D graph showing how the Angular (Rotational) Energy of the system oscillates around a fixed, conserved point over a 50 year time range. I plotted it separate the conservation of linear energy due how it is 3 orders of magnitude more massive.

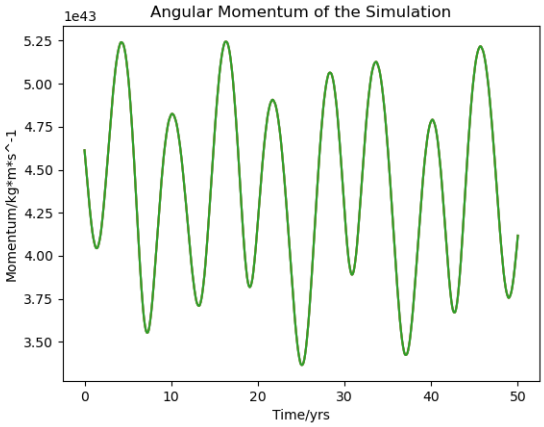


Figure 6: A 2D graph showing how the Angular (Rotational) Momentum of the system oscillates over 50 years. Notice how the angular momentum is several magnitudes more massive than the linear momentum, hence why they are not plotted together.

These graphs are produced each time the simulation runs for the Solar System. They illustrate how energy and momentum, in both the linear and angular forms, are conserved as the simulation progresses. As can be seen in Figures 1 and 2, the simulation accurately demonstrates the orbital paths of the planets in the Solar System, as well as conserving energy as one would expect (in Figures 3 and 5). All the conservation graphs show an oscillatory motion, which is easily explainable as the vectors for each object's position and velocity would follow a sinusodial curve (as each object completes an eliptical orbit), and these vector quantities are used when defining the energy and momentum. The angular energy and momentum have much greater magnitudes than their linear counterparts (due to their physical definitions), and so were plotted on separate graphs to preserve readability.

3 Results and discussion

To evaluate the accuracy of my code, the expected time period of the Earth has been tabulated with each integrating method, and with varying time intervals.This data can be compared to the sidereal year of the Earth - 365.256363004 days[8] - to assess accuracy. The result is shown below:

Time Range of 1 Year:			
	Integrating method:		
Time Step:	Euler Method:	Euler-Cromer:	Verlet Method:
500	364.7251399341648	364.52160473746267	364.7136143118173
1,000	364.78783235836	364.3418250006018	364.71102388304246
10,000	369.01457328374715	364.809179202348	364.69239370663325
50,000	390.907487611304	364.5951935863431	364.94645675975744
100,000	423.5399274092034	364.7009131057543	364.49834616345044
1,000,000	972.0666830939567	376.06580598811695	369.8051541850166

Time Range of 10 Years:			
	Integrating method:		
Time Step:	Euler Method:	Euler-Cromer:	Verlet Method:
5,000	394.6792587841148	319.4053823517646	365.08089758929435
10,000	424.7070972181634	365.2551708338232	364.3106872818726
100,000	695.8507895377325	364.907254069315	364.8163621519103
500,000	1759.0661462344567	367.5369948450219	366.08457519654974
1,000,000	4182.78711664682	378.1410301530341	370.11655209153287
2,500,000	724846.0908558168	443.0992998420084	398.93936799760405

From both tables, it can be observed how increasing the time step dramatically reduces accuracy, regardless of the method used. As anticipated, the Verlet method gives the most consistently accurate results at higher time steps, however its tendency to often underestimate does make the Euler-Cromer method more precise at certain values (at the 100,000 time step in Table 1, and the 10,000 time steps in both tables.). Furthermore, the Euler method, although it does retain accuracy over a 1 year time range and low time steps, is considerably inaccurate both at higher time steps, and over larger time ranges (observe the values generated over a 10 year time range). As explained in more depth in the Background Theory section, the Verlet method has an error one order of magnitude less than the others, hence explaining why for the last row in the 10 year range table it still gives the time period of the Earth as ≈ 399 days, despite many of the orbits "breaking" at such a high time step.

Energy and Momentum conservation

The following data was taken over a 1 year time range, at varying time steps, beginning on the 26th December 2004 (an arbitrary start date). In this data, you can observe how increasing the size of the time steps generally decreases the accuracy of energy conservation, but counter-intuitively appears to actually improve the accuracy of momentum conservation. This is because of the way in which the % error is calculated here.:

% Error of Linear Energy conservation			
	Integrating method:		
Time Step:	Euler Method:	Euler-Cromer:	Verlet Method:
500	0.0565223975663004	0.061141530544057805	0.05930815836392182
1,000	0.05572635194768965	0.06293398939399489	0.05930740391027771
10,000	-0.13002749903752675	0.05979188686458175	0.059290051543612804
50,000	-0.5321987090797063	0.06138648696250553	0.059082312178232295
100,000	-0.8034183734567841	0.06609115717185894	0.058802191183379954
500,000	-2.1261371797618387	0.08897185016851933	0.05934495657058781

% Error of Angular Energy conservation			
	Integrating method:		
Time Step:	Euler Method:	Euler-Cromer:	Verlet Method:
500	0.12180356458343185	0.1218305620319294	0.12164337591419232
1,000	0.12194844459050322	0.12202315003015268	0.12163902958083074
10,000	0.12168296301387194	0.12474677386494447	0.12162693544292005
50,000	0.1266323333079572	0.13708132531172026	0.12153465983565516
100,000	0.1369462453077323	0.1531687705312713	0.12209967099374734
500,000	0.21581707109209072	0.27894312471300736	0.1261384613815047

% Error of Linear Momentum conservation			
	Integrating method:		
Time Step:	Euler Method:	Euler-Cromer:	Verlet Method:
500	-2.569177727026844	-2.5615390632623423	-2.563715565474785
1,000	-2.5781399987169875	-2.5593966260453884	-2.5636170199872477
10,000	-2.7171477640649684	-2.5594446768259145	-2.5623647234313176
50,000	-2.8055931136793206	-2.5484096665486495	-2.5573364914356373
100,000	-2.6282483577397806	-2.5298781803716444	-2.548289205364893
500,000	-2.5738986600396285	-2.407773341057684	-2.523830265143402

% Error of Angular Momentum conservation			
	Integrating method:		
Time Step:	Euler Method:	Euler-Cromer:	Verlet Method:
500	-10.960024603031675	-10.959913726201295	-10.960098613510597
1,000	-10.959735579705434	-10.959570296336576	-10.959922406058949
10,000	-10.963785108545366	-10.955716961206768	-10.955585439025011
50,000	-10.913981869077409	-10.93661065089807	-10.935725046305393
100,000	-10.87546744442105	-10.919657266509153	-10.91810435387325
500,000	-10.494331324744913	-10.789929418811099	-10.77918262896067

For this data, the equation for % error is:

$\sigma = \|\frac{v_A - v_E}{v_E}\| * 100\%$

Where σ is % error, v_A is the actual value observed (at the end of the simulation), and v_E is the expected value (at the start of the simulation).

The problem in the inaccuracy of this data stems from how the energy and momentum constantly oscillate (see the Validation of Methodology section). As a result of this, the value obtained at the end of the simulation is going to be different from that at the start, regardless of the integration method used. Despite this limitation, the data is still of some use. The table showing the % error of Linear Energy conservation verifies that the Verlet method has a much smaller error accumulated for greater iterations, with the % error remaining similar throughout up to 3d.p from the smallest to largest time steps. Also, as expected, the Euler method appears to lose energy at larger time steps, whereas the Euler-Cromer method retains a % error at a middling level of consistency.The table for the % error in Angular Energy conservation similarly shows a higher accuracy in the Verlet method,but also shows the Euler method to be prevalent over the Euler-Cromer method; an error likely stemming from the aforementioned inaccuracy. As the momentum oscillates by a greater relative amount over a 1 year time period, the momentum data has a much higher inaccuracy when compared to the energy data.

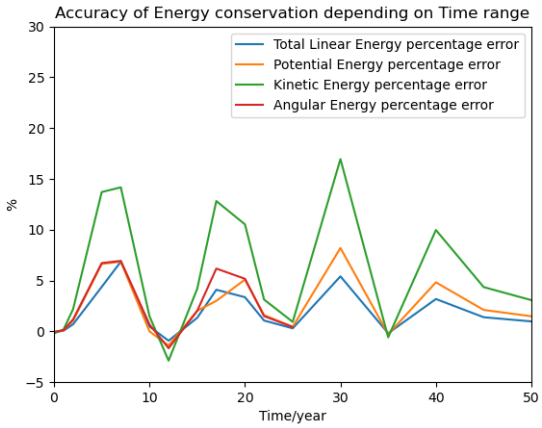


Figure 7: A graph showing how the % error in energy conservation changes depending on time range (from 0.1 to 50 years). Due to the lower number of data points, the accuracy is limited here.

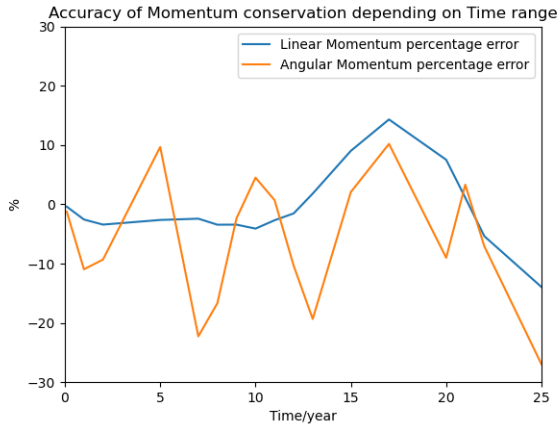


Figure 8: A graph showing how the % error in momentum conservation changes depending on time range (from 0.1 to 25 years). Due to the lower number of data points, the accuracy is limited here.

Figures 7 and 8 show how % errors vary as the time range of the simulation changes (further explaining why the % error in momentum data was so poor). This data was taken by running the simulation, using the Verlet method, over 50,000 iterations beginning on the 26th December 2004. The data shows a clear oscillation in % error depending on the time range of the simulation. These values for the % errors oscillate at the same rate as the values themselves (see Figures 3 - 6), offset by the initial value of the simulation. As a result, the conservation of these qualities was re-attempted between the end result of the simulation over 1 year, and the JPL ephemeris data of the values at the end time. As shown below:

% Error of Linear Energy conservation			
	Integrating method:		
Time Step:	Euler Method:	Euler-Cromer:	Verlet Method:
500	0.000732694965669	0.00388386210535	0.00205155468215
1,000	0.001528696954041	0.00567508519499	0.00205063581694
10,000	0.187209185569	0.00252839866361	0.00202896314738
50,000	0.589786034174	0.00412303430949	0.00181013757924
100,000	0.86279303842	0.00884360894251	0.00152356482335
500,000	2.24249245749	0.0306856935613	0.0016124717158

% Error of Angular Energy conservation			
	Integrating method:		
Time Step:	Euler Method:	Euler-Cromer:	Verlet Method:
500	0.00302831794161	0.00305528336281	0.00286831491571
1,000	0.00316695714376	0.00324157395512	0.00285789172203
10,000	0.00279390210273	0.00585407469558	0.00273634347313
50,000	0.00728804749884	0.0177245846576	0.00215894092973
100,000	0.0170978470946	0.0333009562114	0.00211898800501
500,000	0.0947353479396	0.157785127798	0.00141341756368

% Error of Linear Momentum conservation			
	Integrating method:		
Time Step:	Euler Method:	Euler-Cromer:	Verlet Method:
500	0.00216320590261	0.00567671410014	0.00344285213535
1,000	0.01140589465163	0.00783130203039	0.003499654299
10,000	0.154867595741	0.00698914429824	0.00398745888291
50,000	0.249071524111	0.0148766575373	0.0056184212571
100,000	0.0711461936	0.029807301506	0.0105264390736
500,000	0.0396214885107	0.130825143751	0.00214092367342

% Error of Angular Momentum conservation			
	Integrating method:		
Time Step:	Euler Method:	Euler-Cromer:	Verlet Method:
500	0.000301711275416	0.000426229932257	0.000218582593218
1,000	0.000428078315883	0.000613680800046	0.000218223548532
10,000	0.00768547407739	0.00137287880662	0.00152038062454
50,000	0.0324428202647	0.00696865608742	0.00181013757924
100,000	0.0560507409126	0.00618151824473	0.00792689384063
500,000	0.331143812312	0.00665162707961	0.00543720975976

From this data, the accuracy of the integrating methods can be observed with more ease and all the % errors are drastically reduced. All three methods stay very precise throughout, regardless of time steps, although the Verlet method consistently gives the most accurate results at higher time steps for all the tables. The Euler-Cromer method usually prevails

with accuracy when compared to the Euler method, except at lower time steps when the Euler method occasionally outperforms it. Overall, calculating a % error using the JPL Ephemeris data effectively removes any error due to E / P oscillations, and shows the true precision of the integrating methods.

4 Conclusions

In conclusion, the program can accurately simulate an N-body gravitational simulation, and show how Kepler's 3rd Law, Energy and Momentum conservation, and the accuracy of orbits are conserved. The simulation compared these between three integrating methods and showed that the Verlet method was the most accurate at large time steps and time ranges (giving a % error in Linear Energy conservation of 0.00161 % at the largest time step), that the Euler-Cromer method was effective at smaller time ranges (giving 0.00388 % at $\frac{1}{1000}^{th}$ the time step for the same value), but fluctuated in accuracy at larger time ranges (it gave values of 319.41 and 365.22 days for the time period of the Earth for a range of 10 years and time steps of 5,000 and 10,000 respectively) and for certain conservations (it was often the least accurate approximation for Linear Momentum and Angular Energy conservation), and that the Euler method, although sometimes the most accurate of the three at smaller time steps (it gave values of 0.000733 % and 0.00216 % in Linear Energy and Momentum conservation respectively, better than any other method), quickly became inaccurate at larger time steps and time ranges and was often the least accurate of the three (it gave incredibly inaccurate results for the time range of 1 Earth year at larger time steps). This conclusion is reinforced by the background theory of the methods, which shows that the Verlet method should accumulate a much smaller error for higher numbers of iterations compared to the others, making it preferable for more complex simulations, with the Euler-Cromer method offering a more energy-conserving equivalent to the "basic" Euler method. The program could be further improved by investigating more integrating methods, such as the Runge-Kutta Method or the Euler-Richardson Formula, and comparing the results in the same conditions.Unfortunately, the program also had limitations when determining % errors between the initial and final values in a given range, due to the oscillatory motion of the energies and momenta in the system. These % errors were improved by using JPL Ephemeris data as the expected values in the % error calculations, and showed the considerable impact this had. This highlights the utility of NASA's JPL ephemeris when creating gravitational simulations and further promotes investment in this area, for use in Astrophysics and the growing space sector.

A Additional Information

References

[1] Proposition 75, Theorem 35: p. 956 – Isaac Newton, The Principia: Mathematical Principles of Natural Philosophy. 5th July 1687

[2] Atkinson, Kendall A. (1989). An Introduction to Numerical Analysis (2nd ed.). New York: John Wiley & Sons.

[3] Feynman, Richard P (1963). The Feynman Lectures on Physics, Vol. 1, Sec. 9.6

[4] Cromer, Alan (1981-05-01). "Stable solutions using the Euler approximation". American Journal of Physics. 49 (5): 455–459

[5] Evan Anzalone & Patrick Chai, "Numerical Integration Techniques in Orbital Mechanics Applications", AE 6354 Advanced Orbital Mechanics Georgia Institute of Technology, Atlanta

[6] Hairer, Ernst; Lubich, Christian; Wanner, Gerhard (2003). "Geometric numerical integration illustrated by the Störmer/Verlet method". Acta Numerica. 12: 399–450

[7] Hairer E, Lubich C, Wanner G. "Geometric numerical integration illustrated by the Störmer–Verlet method". Acta Numerica. 2003;12:399-450

[8] "Useful Constants". International Earth rotation and Reference systems Service (IERS). February 13, 2014. Retrieved February 29, 2020.